UNIT II (BEEE)

Machines and Measuring Instruments

IMPORTANT QUESTIONS

1. **What is an Electrical Machine? Briefly explain the conversion of an electrical machine from one form to another form with principle.**

Electrical Machine

- An electrical machine is a device that converts electrical energy into mechanical energy is called Motor.
- An electrical machine is a device that converts mechanical energy into electrical energy is called Generator

Working Principle of DC Generators

An electrical generator is a machine which converts mechanical energy (or power) into electrical energy (or power). whenever a conductor cuts magnetic flux, dynamically induced e.m.f. is produced in it according to Faraday's Laws of Electromagnetic Induction. This e.m.f. causes a current to flow if the conductor circuit is closed. Hence, two basic essential parts of an electrical generator are (i) a magnetic field and (ii) a conductor or conductors which can so move as to cut the flux.

Working Principle of DC Motor

An Electric motor is a machine which converts electric energy into mechanical energy. Its action is based on the principle that when a current-carrying conductor is placed in a magnetic field, it experiences a mechanical force whose direction is given by Fleming's Left-hand Rule and whose magnitude is given by F = BIL newton

Where,

B = magnetic flux density,

I = current and

L = length of the conductor within the magnetic field.

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2. **Compare the Operational difference between Transformer & Induction motor.**

Aspect	**Transformer**	**Induction Motor**
Basic Function	Transfers electrical energy from one circuit to another through electromagnetic induction.	Converts electrical energy into mechanical energy.
Type of Device	Static device (no moving parts).	Rotating machine with moving rotor.
Energy Conversion	Electrical → Electrical	Electrical → Mechanical
Working Principle	Works on the principle of mutual induction between primary and secondary windings.	Works on electromagnetic induction and rotating magnetic field.
Main Parts	Primary winding, Secondary winding, Magnetic core.	Stator, Rotor, Air gap, Shaft, Bearings.

Aspect	Transformer	Induction Motor
Supply	Operates only on AC supply.	Operates mainly on AC supply (single-phase or three-phase).
Output	Electrical power at different voltage/current levels.	Mechanical rotation producing torque and speed.
Speed	No moving parts, so speed is not applicable.	Rotor rotates slightly below synchronous speed.

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3. Classify Different type of Measuring Instruments to measure AC & DC Quantities with basic principle.

◆ **Classification of Measuring Instruments**

Type of Instrument	Principle of Operation	Suitable For	Key Notes
Permanent Magnet Moving Coil (PMMC)	Current through a coil in a permanent magnetic field produces deflection proportional to DC value	DC current & voltage	Highly accurate, linear scale, cannot measure AC directly (requires rectifier)
Moving Iron (MI)	Soft iron piece moves due to magnetic field produced by current in coil	AC & DC current/voltage	Rugged, cheaper, non-linear scale
Electrodynamometer (Dynamometer type)	Interaction between fixed and moving coils carrying current	AC & DC power measurement	Used in wattmeters, accurate but costly
Hot-wire Instrument	Heating effect of current causes expansion of wire, producing deflection	AC & DC current	Slow response, used for RF measurements
Electrostatic Instrument	Force between charged plates produces deflection	AC & DC voltage	Used for high-voltage measurement, very sensitive
Induction Type Instrument	Based on electromagnetic induction (similar to induction motor principle)	AC only (energy meters, frequency meters)	Cannot measure DC, widely used in energy meters

Basic Principles

- **PMMC (DC only)**: Works on **Lorentz force law**. Current in the coil interacts with permanent magnetic field → torque → pointer deflection.
- **MI (AC & DC)**: Works on **magnetic attraction/repulsion**. Iron piece moves depending on strength of magnetic field.
- **Dynamometer**: Based on **mutual inductance** between fixed and moving coils. Torque proportional to product of currents.
- **Hot-wire**: Relies on **Joule heating effect**. Wire expands with heat → mechanical displacement.
- **Electrostatic**: Uses **Coulomb's law**. Electrostatic force between charged plates causes deflection.
- **Induction type**: Works on **Faraday's law of electromagnetic induction**. Alternating flux induces eddy currents → torque.

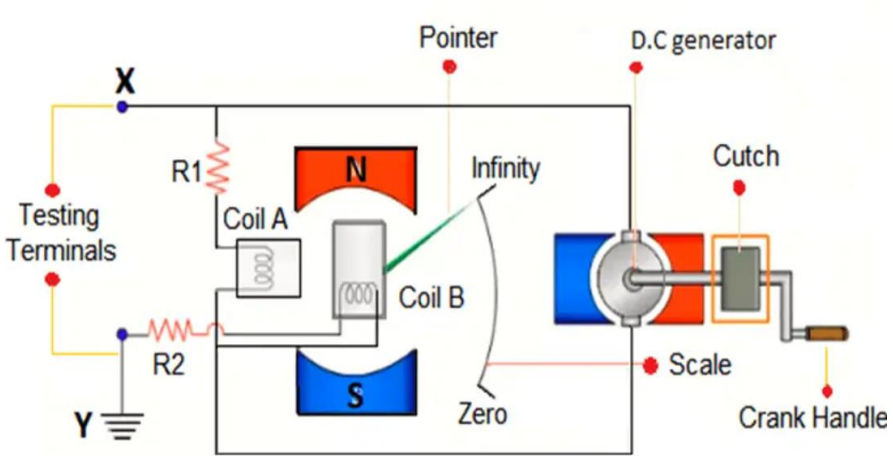
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4. Classify different Resistors? How to Measure High Resistance? Explain with a neat sketch.

a. Low resistance – we can find by using ohm's law
b. Medium resistance – we can find by using Wheatstone bridge
c. High resistance – we can find by using megger

Megger is used to measure a high value of resistance. Megger consists of the following parts.

1. DC generator
2. 2 Coils (Coil A, Coil B)
3. Clutch
4. Crank handle
5. terminal X & Y



- The megger consists of a DC generator; its armature is hand-driven to generate a voltage. A clutch mechanism is used to slip after reaching a certain determined speed.
- Resistance R1, R2 connected in series with two coils, Coil A and Coil B which constitute one instrument to which an indicating pointer is attached.
- Permanent magnets having North, South poles produce a magnetic field to deflect the pointer. Test terminals at X and Y are used to measure insulation resistance.

- The megger is designed in such a way that the needle floats freely till the generator is operated. When the generator is not operating, the needle may come to rest at any point on its scale.
- The megger is used to measure a high quantity of insulation resistance. The high resistance may be between the windings of a transformer or motor or between the conductor in a cable and the conduit or sheath enclosing the cable.
- If the test leads connected to the line and earth terminals are open-circuited and a hand-cranked generator is operated, the needle moves to infinity. Infinity resistance means that it is too high for the instrument to measure.
- If the test leads are connected to each other while the hand crank is turned, the pointer will defect to zero, indicating no resistance between the test leads.
- A zero deflection in the above-mentioned test can mean that the conductor under test is touching the sheath or conduit surround it.

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5. What is the Difference between Back emf & Generated emf? State the function of each & Explain Briefly

Difference between Back EMF and Generated EMF

Aspect	**Back EMF**	**Generated EMF**
Definition	Back EMF is the electromotive force induced in the armature of a DC motor that opposes the applied voltage.	Generated EMF is the electromotive force produced in the armature of a generator due to rotation in a magnetic field.
Occurrence	Occurs in **motors**.	Occurs in **generators**.
Direction	Opposes the applied voltage according to **Lenz's law**.	Drives current in the external circuit.
Purpose	Limits the armature current and protects the motor from excessive current.	Produces electrical power from mechanical energy.
Energy Conversion	Electrical → Mechanical.	Mechanical → Electrical.
Formula	$Eb = V - IaRa$	$Eg = (P\Phi ZN) / (60A)$

Function of Back EMF

1. Controls armature current **in the motor.**
2. Protects the motor from damage **due to excessive current.**
3. Helps maintain stable motor speed**.**

Explanation:
When a DC motor runs, the rotating armature cuts magnetic flux and an EMF is induced. According to Lenz's law, this EMF opposes the applied voltage, so it is called **Back EMF**.

Function of Generated EMF

1. **Produces electrical energy** in a generator.
2. **Supplies current to the external circuit**.
3. **Enables energy conversion from mechanical to electrical**.

Explanation:
In a generator, when the armature rotates in a magnetic field, conductors cut magnetic flux and an EMF is induced. This EMF drives current through the external load and is called **Generated EMF**.

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6. **Why Single Phase Induction Motors are not Self Started? Explain Any two types of Single Phase Induction Motors with neat diagrams.**

A **single-phase induction motor** is **not self-starting** because a single-phase AC supply produces only a **pulsating magnetic field**, not a rotating magnetic field.

When the stator is energized with single-phase AC, the magnetic field alternates in magnitude but **does not rotate**. According to the **double revolving field theory**, this pulsating field can be resolved into **two equal rotating magnetic fields** that rotate in **opposite directions** at synchronous speed.

Types of Single-Phase Induction Motor

- Split-phase motor
- Capacitor-start motor
- Capacitor-start capacitor-run motor (or two-value capacitor motor)
- Permanent-split capacitor (PSC) motor (or single-value capacitor motor)
- Shaded-pole motor

1. **Capacitor Start Induction Motor**

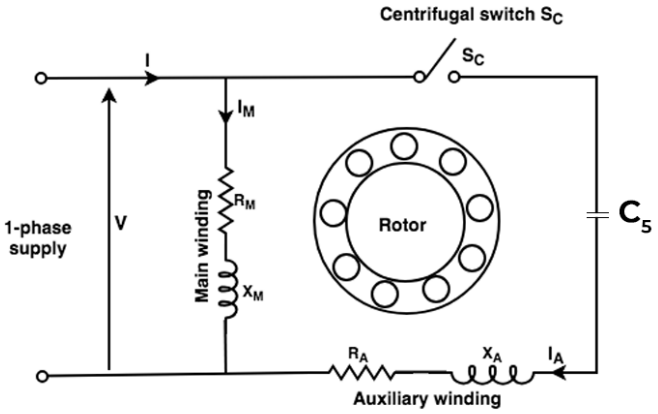
It consists of:

- Main winding
- Auxiliary winding
- **Capacitor connected in series with the auxiliary winding**

Working Principle

- The capacitor creates a **larger phase difference (about 90°)** between the currents in the two windings.
- This produces a **strong rotating magnetic field** and **high starting torque**.

Once the motor reaches about **75% speed**, the **centrifugal switch disconnects the capacitor and starting winding**



Capacitor-Start Motor

2. **Capacitor-Start Capacitor-Run Motor (Two-Value Capacitor Motor)**

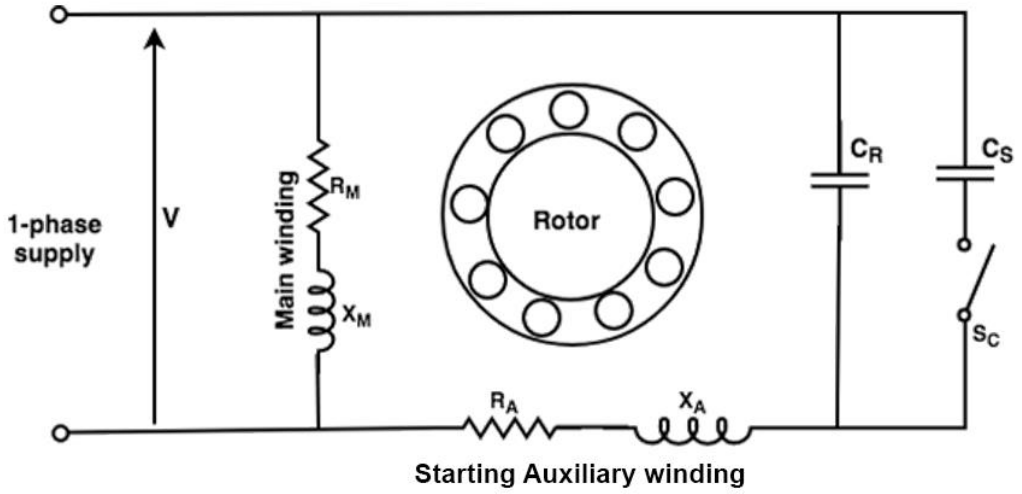
A **Capacitor-Start Capacitor-Run Motor** is a type of **single-phase induction motor** that uses **two capacitors**:

1. **Starting Capacitor (Cs)**
2. **Running Capacitor (Cr)**

These capacitors are connected with the **auxiliary (starting) winding** to produce a **phase difference** between the currents of the main winding and auxiliary winding.

Operation

1. **At Starting**
   - Both **starting capacitor (Cs)** and **running capacitor (Cr)** are connected in the circuit.
   - This combination gives **large capacitance**.
   - The large capacitance produces a **phase difference close to 90°** between the currents in the main and auxiliary windings.
   - As a result, a **strong rotating magnetic field** is produced.
   - This provides **high starting torque** to start the motor.
2. **After Starting**
   - When the motor reaches about **70–80% of its rated speed**, the **starting capacitor (Cs)** is disconnected by a **centrifugal switch**.
   - Only the **running capacitor (Cr)** remains in the circuit.
3. **During Running**
   - The running capacitor maintains a **good phase difference** between the windings.
   - This improves **power factor, efficiency, and smooth operation** of the motor.



Capacitor-Start Capacitor-Run Motor (Two Value Capacitor Motor)

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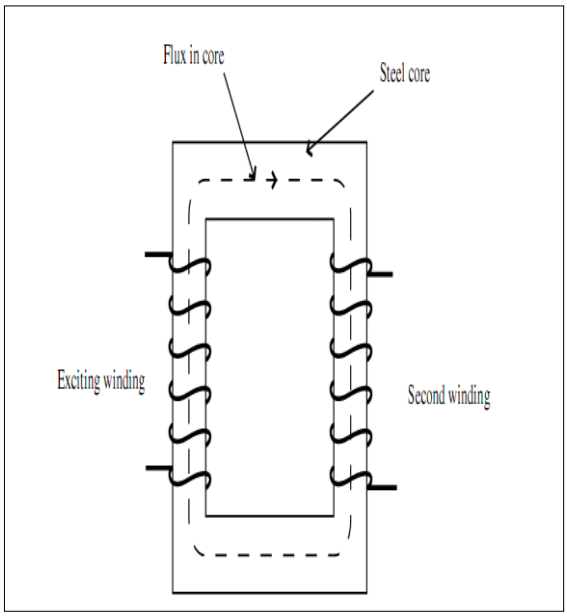
7. **What is the Role of Transformer in Electrical Power System? Explain the operation of Transformer with neat diagram.**

Transformer

A transformer is a stationary electric machine which transfers electrical energy (power) from one circuit to another circuit with the same frequency but with different voltages and currents.

Working Principle of Transformers

- The transformer works on the principle of Mutual induction between two magnetically coupled coils and it is based on Faraday's laws of Electromagnetic Induction
- When primary winding is connected to AC supply mains a current flow through it and this current produces an alternating flux in the core
- This flux links with primary winding and produces self-induced e.m.f in the primary winding which opposes the applied voltage
- This flux passes through the core and links with the secondary windings to induce an E.M.F called mutually induced E.M.F in the secondary winding
- The frequency of the emf induced in the secondary is same as that of the flux or that of the supply voltage.
- The induced emf in the secondary winding will be able to circulate the current in the external load connected to it
- Energy is transferred from primary winding to the secondary winding by electro-magnetic induction principle without change of frequency



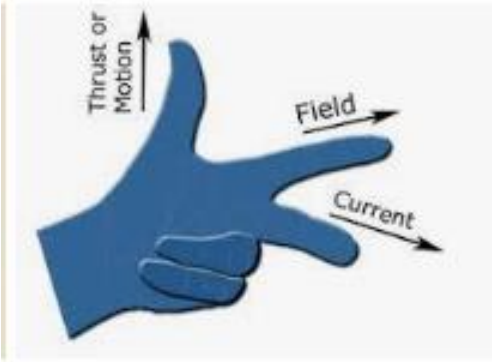
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8. How to Find the direction of Mechanical & Electrical output of the Machines & Explain Principle of each.

Fleming's left-hand rule

Fleming's left-hand rule is a physics principle that determines the direction of force on a current-carrying conductor in a magnetic field:

- Arrange your left hand so that your thumb, forefinger, and middle finger are perpendicular to each other.
- Point your forefinger in the direction of the magnetic field.
- Point your middle finger in the direction of the electric current.
- Your thumb will point in the direction of the force on the conductor.

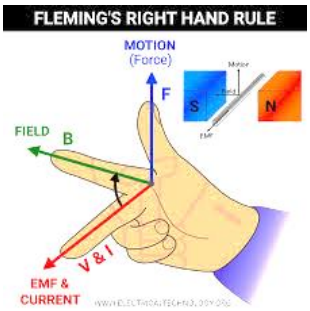


Fleming's Right-hand rule

Fleming's right-hand rule is a rule in electromagnetism that shows the direction of induced current when a conductor moves in a magnetic field. It can be used to determine the direction of current in a generator's windings. To use Fleming's right-hand rule, you:

- Hold your right hand out with your thumb, index finger, and middle finger perpendicular to each other
- The thumb points in the direction of the motion or applied force

- The index finger points in the direction of the magnetic field
- The middle finger points in the direction of the induced current



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9. Describe the Role of Commutator in DC Machines? Explain.

- The basic purpose of the commutator is to convert AC to DC. It functions as a reversing switch, and its purpose in the generator is explained below.
- The electromagnetic field produced within the generator's armature coil alternates, leading to an alternating current flowing within the armature coil.
- This current may be reversed through the commutator at the precise moment that the armature coil crosses the magnetic unbiased axis. As a result, the load develops a DC or unidirectional current.
- The commutator ensures that the current flowing from the generator continues in the same direction indefinitely. By moving along the commutator, the brushes create high-quality electrical connections between the generator and the load.

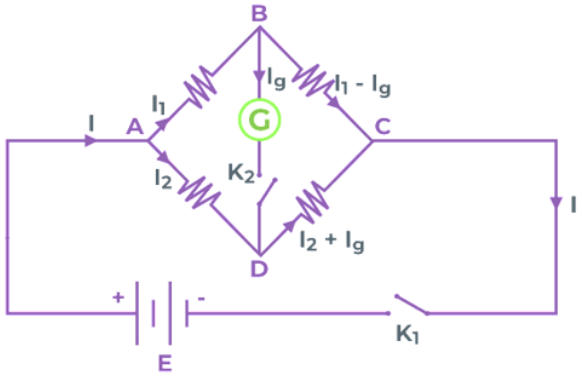
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10. How to Find the Medium Resistor? Explain the principle & Derive the expression for the unknown Resistor.

- Wheatstone Bridge works on the principle of null deflection i.e., there is no current flowing through the galvanometer, and its needle shows no deflection, hence the name null deflection.
- In the unbalanced state of the Wheatstone bridge i.e., when the potential across the galvanometer is different, the galvanometer shows the deflection, and as the bridge becomes balanced by changing the variable resistor, the potential difference across the galvanometer becomes zero i.e., the equilibrium state of Wheatstone bridge.
- Construction of Wheatstone Bridge requires four resistors P, Q, R, and S that are placed in the form of four sides AB, BC, AD, and DC of a quadrilateral ABCD. A cell E and key K1 are placed between the A and C ends of this quadrilateral, and a sensitive galvanometer G and key K2 is placed between the B and D ends. Clearly, the potential of point A will be equal to the potential of the positive plate of the cell and the potential of point C will be equal to the potential of the negative plate of the cell.
- When the bridge is in an equilibrium state, that is, there is no deflection in the galvanometer. That is, in the equilibrium state of the bridge, the ratio of the resistances of any two adjacent arms is equal to the ratio of the remaining two adjacent sides.
- Suppose, on pressing the cell key K_1, a current I flows through the cell, which splits into two parts at the end A.

- One part I_1 flows through the resistance P in arm AB and the other part I_2, through the resistance R in arm AD.
- The current I_1 again comes to end B and gets divided into two parts.
- One part of it Ig flows through the galvanometer in arm BD and the remaining part ($I_1 – I_g$) flows through resistance Q in arm BC.
- At the end D, the current I_2 from arm AD and the current I_g from arm BD, so the current flowing through the resistance S in arm DC will be ($I_2 + I_g$).
- So according to Kirchhoff's law, in closed path ABDA,
- $I_1P + I_gG – I_2R = 0$ . . . (1)
- And in closed path BCDB,
- $(I_1 – I_g)Q – (I_2 + I_g)S – I_gG = 0$ . . . (2)
- The values of the resistors P, Q, R, and S are taken in such a way that no current flows through the galvanometer G when the key K_2, is pressed.
- This is called the equilibrium state of the bridge, that is, in the equilibrium state of the bridge, the deflection in the galvanometer is zero ($I_g = 0$).
- Putting Ig = 0, in the above equations,
- $I_1P = I_2R$ and $I_1Q = I_2S$ or $I_1P / I_1Q = I_2R / I_2S$ or $P / Q = R / S$
- This is the necessary condition for the balance of the Wheatstone Bridge. With the help of the above formula, knowing the values of three resistors P, Q, and R, the value of the fourth resistance S can be found.
- **Wheatstone Bridge Formula**
- The Wheatstone Bridge Formula for the calculation of the unknown resistor is as follows:
- R=P/S * Q OR R = PS /Q
- where,

  P and **Q** is the resistance of ratio arm **S** is the known resistance of the standard arm **R** is the unknown resistance.



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11. Draw the neat sketch the PMMC ? Explain the Operation.

Working: When the instrument is connected in the circuit, operating current flows through the coil, which is mounted on the spindle. Since the coil is placed in the strong magnetic field, a force acts on the coil which produces deflecting torque (T_d). Thus pointer attached to the spindle deflected over the scale. If the current in the coil is reversed, the direction of the

deflecting torque will be reversed, because field produced by the permanent magnet does not change. This change will give wrong direction of rotation. So this instrument can not be used for AC supply

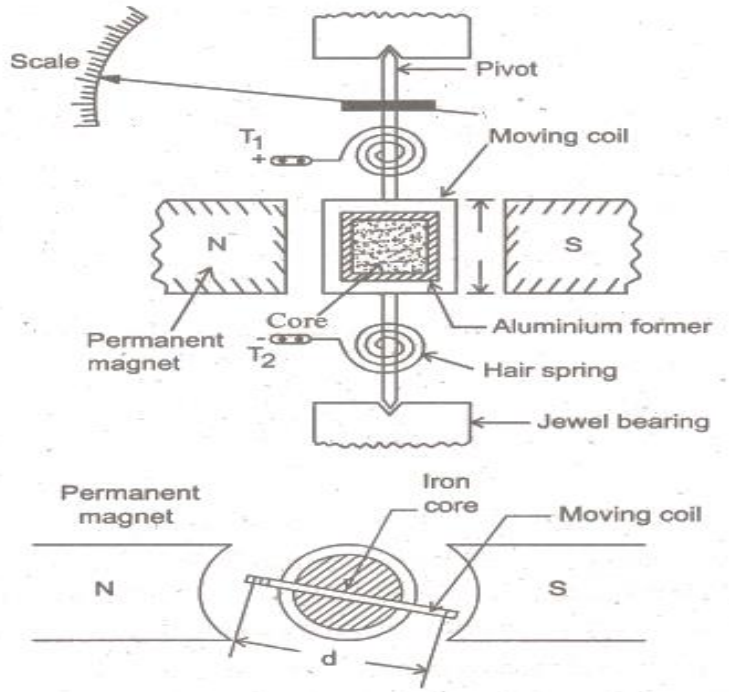


Figure 5.6 PMMC

Deflecting torque (T_d) :

When the current is passed through the moving coil, the instrument is being used as an ammeter. When this current carrying coil is placed in magnetic field, it will experience a mechanical force.

The force exerted on each conductor

F=BIL Newton

B= flux density in wb/m^2

L= length of the coil in meters

I= Current to be measured in amperes

Let N= Number of conductors ,then force acting each coil side

F=BILN Newton

Deflecting torque (T_d) :

The force acting in opposite direction on the two coil sides gives rise to a torque

therefore, Deflecting torque , T_d =F x b

Where b= breath of the coil

T_d = BILN b

= BIN (L x b) [a= L x b]

= BINA

= BNA I

= GI [BNA= G= constant]

T_d= GI

The Controlling torque is given by

$T_c = K_s \theta$

Under equilibrium condition, $Tc = T_d$

$K_s \theta = GI$

$\theta = \frac{G}{K_s} I$

$\theta \propto I$

Therefore, the deflection of the pointer is directly proportional to the current measured. Hence such instruments have uniform scale.

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12. Compare Difference type of MI types of Instruments & Explain their Principles.

Types of MI Instruments

Type	Principle	Construction	Operation	Advantages	Limitations
Attraction Type	A soft iron piece is attracted into the coil when current flows, due to magnetic field strength	Coil with a pivoted soft iron vane inside	Current produces flux → iron piece is pulled into coil → pointer deflects	Simple design, rugged	Scale is non-linear, less sensitive
Repulsion Type	Two soft iron pieces are magnetized with like polarity, causing repulsion	One fixed iron piece inside coil, one movable iron piece attached to pointer	Current produces flux → both iron pieces magnetized → repulsion → pointer deflects	More torque, better sensitivity than attraction type	Still non-linear scale, friction errors possible

Principles

1. **Attraction Type MI Instrument**
   - When current flows through the coil, it produces a magnetic field.
   - A soft iron piece inside the coil gets magnetized and is **attracted** into the stronger magnetic field region.
   - This movement causes the pointer to deflect on the scale.
   - Deflection ∝ square of current (hence works for both AC & DC).
2. **Repulsion Type MI Instrument**
   - Two iron pieces are used: one fixed, one movable.
   - When current flows, both pieces are magnetized with the same polarity.
   - They **repel** each other, and the movable piece pushes the pointer.
   - Deflection again ∝ square of current (works for AC & DC).

BEST OF LUCK